What Survives Distillation? A Pre-registered Study of Training-Free Acceleration for Few-Step and Autoregressive Video Diffusion

Anonymous CVPR submission

Abstract

Video diffusion models are deployed as distilled few-step and autoregressive students, yet the training-free accelerators paired with them were designed on 50-step teachers. We measure what survives on six released students in three families, each against its own teacher, and test pre-registered predictions on a held-out family, scale, prompts and GPU. Three conclusions follow. First, distillation changes the sampling grid and the system, not the features: at matched noise level a student's attention is as concentrated as its teacher's, on the same states its features are closer to its teacher's than to its own one step away, and the wider interval each step covers accounts for its lost redundancy, while a few wide steps amplify early errors and the VAE decoder and attention kernel cap the gain. Second, calibrations made on the teacher therefore mislead: cache signals mis-rank where a student can skip, and block-sparse attention over position blocks loses to dropping a step at equal speed, even with an oracle choice of blocks. A latency probe and an 8-prompt sweep on the student replace them, and they show that a skip's fidelity cost and its quality effect can disagree. Third, acting on both, two calls on the grid, a tiny decoder and compile reach 2.5–3.5× end to end on four students, within 1% of the probe's prediction and at stated quality costs; we offer this as a baseline for future accelerators. We release the probe, the protocol and the registration record.

1. Introduction

Video diffusion models are increasingly deployed as distilled *students*: few-step bidirectional models that need 4–8 network evaluations instead of 50 [5, 43], and autoregressive (AR) models that stream a video chunk by chunk [7, 37]. The training-free accelerators paired with them were designed, tuned and evaluated on 30–50-step teachers. These include step caches that reuse features across denoising steps [12, 15, 19], sparse attention [28, 40] and post-training quantisation [9]. Whether they still pay off after distillation is mostly asserted rather than measured. Several works state that step caching degrades on few-step models [13, 24, 46], because they "have no inter-step redundancy left" [38], or that video attention is not reliably sparse [1]. Both statements motivate *trained* replacements.

We ask what survives distillation, and why, in a controlled setting. We study three families: Wan2.1 [25], LTX-Video [6] and FLUX.2-klein, each with its teacher and its released students (two bidirectional, two AR, one image and one held-out video student). After studying the Wan family, we *pre-registered* predictions for a held-out family, a held-out scale (rCM on Wan2.1-14B with its teacher), held-out prompts and a second GPU, and we report every outcome, including the predictions that failed.

Three conclusions. *The grid and the system change, not the features.* At matched noise level a student's attention is as concentrated as its teacher's. On the same states the teacher's features change as much as the student's across a step, and a student's features are closer to its teacher's than to its own one step away. What changes is the grid, a few wide steps that amplify early errors, and the system: with 4–7 DiT calls instead of 50, the VAE decoder and the attention kernel's speed at the model's own shape cap what any DiT accelerator can deliver.

So teacher calibrations mislead, and a short measurement on the student does not. Cache signals mis-rank where a student can skip, and block-sparse attention over position blocks loses to dropping a step at equal speed, even with an oracle choice of blocks. An 8-prompt sweep and a latency probe, taken on the student in minutes, find the skip positions and predict the speedups, and they show that a skip's fidelity cost and its quality effect can disagree.

Acting on both gives a baseline to beat. Two calls on the grid, a tiny decoder and compile reach 2.5–3.5× end to end on four students, within 1% of the prediction and at stated costs. Allocating compute across calls and frame-interleaved sparsity did not beat the grid at equal speed. Table 1 summarises what transfers and what decides it.

Contributions. (i) A controlled, teacher-matched measurement of six released students in three families. (ii) The three conclusions above, tested by pre-registered predictions with all outcomes reported. (iii) A protocol for measuring acceleration on students, a latency-split probe with an

Accelerator (designed on teachers)	On students	Evidence	Decided by
Step cache with a teacher-calibrated signal (FBCache, TeaCache)	mis-ranks	tuned on the student to skip one call, they never beat the measured skip and are worse in fidelity or quality for 4 of 5 student–cache pairs; FBCache's signal orders 6 of 15 LTX pairs correctly (Sec. 5.1)	grid: amplification, sampler
One skipped call at a measured position	transfers	1.16–1.29×; 8 tuning prompts find the cheapest position (22/23 pairs); confirmed on held-out prompts and at 14B (Sec. 5.4)	measurement
Training-free block-sparse attention (pooled top-k over position blocks)	loses	drifts further than dropping a step at equal speed (+0.08 to +0.18 LPIPS), even with an oracle choice of blocks; $\leq 1.05\times$ on LTX; a semantic-aware port matches it in drift under an ideal kernel (Sec. 5.2)	selector, kernel shape, GPU
INT8 W8A8, compiled	varies	DiT $\leq 1.04\times$ at 1.3B, $1.13\times$ at 14B, $1.23\times$ at 13B, teacher and student alike (Sec. 4.3)	model size, not distillation
More steps on the first AR chunk	transfers	one extra call more than halves the drift of a reduced schedule (Sec. 5.3)	context anchoring
Any DiT accelerator	capped	the VAE decoder takes 26–35% of a 1.3B Wan student's latency, 9.5% at 14B, 39–47% on a B200 (Sec. 4.3)	latency split
Two calls + tiny decoder + compile	baseline	2.53–3.45× on four students, within 1.0% of the probe; drift within 14% of two calls alone; against the base, motion falls on all four, DINO on two, aesthetics on one (Sec. 6)	grid and system

Table 1. What transfers from teacher to student, and what decides it (end-to-end speedups on one A800; last row, our baseline, on a PRO6000 BW).

8-prompt skip sweep, and a baseline it predicts.

2. Related Work

Few-step and autoregressive students. Video generators are now deployed as distilled students. Distillation by distribution matching [35, 36], by continuous-time consistency, alone or with a DMD regulariser [16, 43], and by on-policy flow maps [5] brings Wan2.1 [25] to 1–4 network evaluations. Autoregressive (AR) students [7, 32, 37] generate chunk by chunk from a clean-context key–value (KV) cache and target streaming latency. Vendors also release official step-distilled checkpoints such as LTX-Video [6].

Training-free acceleration, designed on many-step teachers. Three families dominate, each tuned on 30–50-step samplers. They are step caching and forecasting [2, 12, 14, 15, 19, 22, 44], sparse attention [10, 28, 33, 40, 42] (including trainable variants [39]), and post-training quantisation [3, 9]. Step caches decide when to skip from a cheap signal calibrated on the teacher's trajectory. TeaCache [12] uses the change of a timestep-modulated input, First-Block Cache (FBCache)[1] the change of the first block's residual, and MagCache [19] residual magnitude ratios. A calibrated schedule is tied to the sampler, step count, resolution and precision it was calibrated with [27], and the best recipe differs across models and hardware [11].

[1] https://github.com/chengzeyi/ParaAttention

Acceleration of few-step and AR students. Several recent works assert that step-wise reuse stops working after distillation [13, 24, 34, 38, 46]. For attention, the real-time AR method MonarchRT finds that video attention "is not reliably sparse" [1]. The responses are mostly trained: learnable caches [46], block-sparse attention distilled jointly with steps [4], and a fully trained stack of step distillation, sparse-linear attention and 8-bit weights and activations (W8A8) [41]. For AR students, chunk-aware sparsity allocation [17] and trainable persistent-block sparsity [29] report that attention concentrates on a persistent subset of the context, and chunk-wise caches [18] and, concurrently, cache anchors [30] treat chunks differently. Training-free work on students includes frame-interleaved sparsity [24], solver-aware cache targets [34], routing between large and small models at calibrated steps [21], progressive resolution [26], head-specific KV caches [23], and fewer steps for later AR chunks plus runtime engineering [20]. None of these works measures *why* an accelerator transfers or not; we do, against each student's own teacher.

Measuring acceleration. Reported speedups often mix bases (FLOPs vs latency, DiT-only vs end-to-end, different GPUs for quality and speed). Caches are scored by fidelity to

Family	Model	Steps	CFG	Output
Wan2.1-1.3B	teacher	50	5.0	480×832×81
	AnyFlow, rCM	4	–	
	AnyFlow-FAR, Self-Forcing (AR)	4/chunk	–	
Wan2.1-14B	teacher (probes only)	50	5.0	as 1.3B
	rCM (held-out scale)	4	–	
LTX-Video 13B	dev	30	3.0	512×704×121
	distilled	7	–	
FLUX.2-klein 4B	base	50	4.0	1024^2 image
	distilled	4	–	

Table 2. Teacher–student pairs, run with their released sampling settings (CFG: classifier-free guidance scale).

the uncached output (LPIPS, PSNR, SSIM) [14], which on students can disagree with reference-free quality (Sec. 5.1). Once the DiT is cheap, the VAE decoder becomes the bottleneck [45]; we quantify this per model.

3. Setting and Protocol

Teacher–student pairs. We study three families, each with a teacher and the students released for it (Tab. 2). Wan2.1-1.3B [25] comes with two bidirectional students, AnyFlow [5] and rCM [43] (public checkpoints, at both scales, marked as an unofficial reproduction), and two AR students, AnyFlow-FAR (frame-autoregressive) and Self-Forcing [7]. LTX-Video 0.9.7 (13B) [6] has official *dev* and *distilled* checkpoints, and FLUX.2-klein (4B) gives an image pair, to test whether the findings are specific to video. Every model runs with its official or model-card sampling settings, including VAE precision and decode settings. For LTX we run the single-scale first pass, without the optional latent upsampler and refinement pass. Skip and quality experiments use AnyFlow, rCM, klein and LTX; the AR students enter the probes, the AR schedules (Sec. 5.3) and the latency split. LTX-Video is the held-out family: its predictions were registered before any LTX student run (Sec. 7). rCM on Wan2.1-14B, a held-out scale registered the same way, shares the 1.3B rCM's sampler, VAE and output size; its teacher enters the probes.

Prompts and seeds. Development uses 33 VBench [8] prompts stratified over its dimensions (seed 0). The held-out set is 33 disjoint VBench prompts (seeds 0 and 1), and images use 64 DrawBench prompts. Eight separate tuning prompts per modality serve only probes and calibration.

Accelerators. All accelerators are training-free and plug into the released pipeline. On the step axis we use residual reuse of the whole block stack at a chosen call and first-order Taylor forecasting [15]. On teachers we also use FBCache, TeaCache [12] and MagCache [19] with their published calibrations. On the spatial axis we use block-sparse self-attention with pooled-score selection [31, 40]. It keeps a fraction d of 128×128 blocks, runs on a FlexAttention kernel, and keeps the first w steps dense. System-level options are W8A8 dynamic INT8 (torchao), `torch.compile` of every block, and VAE precision.

Speed. All timings run on one NVIDIA A800 80GB PCIe GPU (14B: a second one) with 1–2 warm-up and 3–5 timed runs, and a run is discarded if another process appears on the GPU. The A800 servers were later reclaimed, so the system axis is repeated on a B200 and later experiments ran on B200s, H100s, a PRO6000 BW and, for drift only, 4090s and A100s; the supplement lists which, and each comparison stays on one GPU type. We report end-to-end time split into text encoder, DiT, VAE and other, and peak memory. Every speedup is a ratio of runs on the same GPU. Timings use PyTorch's default numerics (cuDNN convolutions in TF32) and quality runs disable TF32, which matters only for fp32 decoders (supplement). A latency-split probe adds CUDA-event timing of self- and cross-attention, and the sparse kernel's speedup at each model's attention shape, to predict an accelerator's effect before it is run (Sec. 4.3).

Quality. *Divergence* from the unaccelerated output of the same model, prompt and seed is measured by LPIPS, PSNR and SSIM. Reference-free metrics are CLIP text alignment, LAION aesthetics, DINOv2 subject consistency, RAFT motion and VBench's imaging quality, motion smoothness and temporal flickering. Differences are paired, with 95% bootstrap confidence intervals over prompts.

Noise floor. Compiling the DiT changes no arithmetic intent, but its rounding noise is amplified along the trajectory. The LPIPS between compiled and eager outputs is each model's *compile floor*: AnyFlow 0.027, LTX dev 0.025, LTX distilled 0.075, Wan 0.099, rCM 0.145. Below its floor, a divergence is as small as a change of rounding: we read it as free and do not rank such divergences.

Pre-registration. Predictions, thresholds and decision rules were committed to the project repository before the runs they concern (not third-party timestamped; the supplement lists every commit). Section 7 reports every outcome, including failures.

4. The Grid and the System Change, Not the Features

4.1. Features at matched noise level

Attention is as concentrated. For 256 sampled queries in every fourth layer we compute the exact softmax. The *oracle coverage* at density p is the attention mass held by the top p fraction of keys, the best any selector can reach at

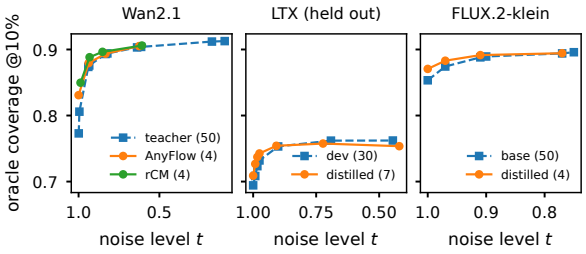


Figure 1. Oracle attention coverage at 10% of the keys vs noise level. Teacher and students coincide at matched noise level in all three families, including the held-out pair (pre-registered).

Interval (t)	Wan teacher		AnyFlow	rCM	FAR, chunks 1–7	14B	
	one step	same interval				teacher	rCM
$1000 \to 940$	0.99	0.51	0.48	0.42	0.76–0.90	0.59	0.31
$940 \to 833$	0.99	0.81	0.75	0.57	0.88–0.93	0.75	0.46
$833 \to 620$	0.995	0.86	0.80	0.63	0.88–0.92	0.70	0.55

Table 3. Block-residual cosine between consecutive calls. Over the same noise interval, the teacher's features change about as much as a student's do in one step. 14B: teacher over the same interval (one step: 0.98).

that density. Teacher and student coincide at matched noise level (Fig. 1): within 0.015 in the Wan family, 0.014 at 14B and, as predicted before any run, 0.018 on the held-out LTX pair; other query samples move coverage by at most 0.002. The one exception is pure noise, where the students are *more* concentrated (0.83–0.85 vs 0.77). Coverage is a fraction of the keys, so we compare it only at equal key count: across families and AR chunks it disagrees with the mass of a fixed number of keys. AR context adds keys, not concentration: in AnyFlow-FAR coverage rises from 0.73 in the first chunk to 0.89 in the last, while the mass of a fixed 256 keys falls from 0.80 to 0.71. At the oracle level, this does not support MonarchRT's premise that video attention "is not reliably sparse" [1], and agrees with the persistent concentration that AR sparse methods build on [17, 29].

The redundancy lives in the grid. We take the cosine between a block's residual (output minus input) at two consecutive calls, averaged over blocks and 4 tuning prompts. For the teacher (conditional branch) we measure it between consecutive steps and over the same noise interval as one student step, between the teacher calls nearest to the student's two noise levels. Between consecutive steps the Wan teacher is almost static (Tab. 3). Over a student-sized interval it changes about as much as AnyFlow does in one step (0.51–0.86 vs 0.48–0.80). Distillation removes the fine grid, and with it the redundancy that caches exploit. We summarise this with a descriptive share, $\sum_k (1 - \cos_k^{\text{teacher}}) / \sum_k (1 - \cos_k^{\text{student}})$: the interval accounts for 84% of the loss on AnyFlow and 55–59% on rCM (both scales) and klein. On these separate trajectories the unexplained gap is at most 0.06 on AnyFlow and 0.017 on LTX, but reaches 0.24 on rCM (0.29 at 14B) and 0.21 on klein. Interval width alone is not the driver: LTX's tiny early steps are not redundant, against our prediction ($\cos \geq 0.90$; measured 0.69–0.86), because its schedule places its smallest steps where features change fastest. There is no block-level refuge either: only the first two blocks and the last stay redundant, capping block-level reuse at about 10%. AR context does raise redundancy: the first chunk behaves like a bidirectional student (0.64–0.78 in AnyFlow-FAR, 0.41–0.57 in Self-Forcing), and later chunks rise to 0.88–0.93 and 0.65–0.80.

On the same states the gap disappears. Those comparisons follow each model on its own trajectory. A registered probe removes that difference: while a student samples, its teacher runs on the states the student sees (8 tuning prompts; AnyFlow, rCM at both scales and LTX, not klein). Over the same two states the teacher's residuals change at least as much as the student's: the share rises to 102–109%, on rCM from 55–59%, so the student-specific loss above came from the differing states. And on one state a student's residuals are closer to its teacher's than to its own one step away: mean cosine 0.79–0.91 vs 0.46–0.75 after the first call, paired difference $+0.10$ to $+0.45$ with all CIs above 0 (one exception by call, LTX's fourth; at the first call, teacher–student 0.62–0.77). Distillation moves a block's residual less than one step of the student does.

4.2. The grid amplifies early errors

A registered diagnostic on the LTX student separates three explanations of why its late skips cost little fidelity. *The decoder* is rejected: with a plain decode, the ratio of last-skip to first-skip cost stays at 0.045 (0.034 as released). *The local error* is rejected too: the error that reuse injects at call k, $|\Delta\sigma_k| \, \|v_k - v_{k-1}\| / \|x_k\|$, *grows* 29× from call 1 to call 6, because late intervals are wide. *Amplification* is supported: a Gaussian state error of equal relative size ($\epsilon = 0.1$ of the state's norm) moves the output by LPIPS 0.114 at call 1, 0.067, 0.014, then 0.003 and about 0.001 at the last three calls. It is not specific to LTX. In a registered test on four more students, an error at the first call moves the output 3.5× (AnyFlow), 6.2× (rCM), 7.2× (rCM-14B) and 10× (Wan2.2) as far as at the last, with first-call values above each compile floor and last-call values below it. The ratios shrink at $\epsilon = 0.3$ (2–3×; LTX 25×), and on the Wan2.1 students the last call rises again, as noise there stays visible. On such a student, fidelity measures how far the trajectory moved rather than whether the video got worse. Block similarity measures the local change and is blind to amplification, which is why it cannot rank skip positions.

Model	e2e (s)	VAE	self-attn	sparse kern.	INT8
Wan teacher (50)	192	3.0%	59%	3.10×	1.03×
AnyFlow (4)	12.8	31.1%	33%	3.08×	1.04×
rCM (4)	11.4	35.3%	39%	3.10×	1.04×
rCM-14B (4)	43.4	9.5%	46%	3.15×	1.13×
Self-Forcing (AR)	12.4	32.4%	30%	0.40×	0.95×
klein (4)	1.16	13.8%	13%	0.58×	–
LTX dev (30)	59.3	0.9%	15%	1.00×	1.24×
LTX distilled (7)	7.77	6.8%	14%	1.08×	1.23×

Table 4. Latency split as released, one A800 per measurement (PyTorch default numerics): e2e time and VAE share from timing runs; self-attention share and sparse kernel speedup (density 0.2, own shape) from the latency probe. INT8: DiT speedup of compiled W8A8 over compiled bf16.

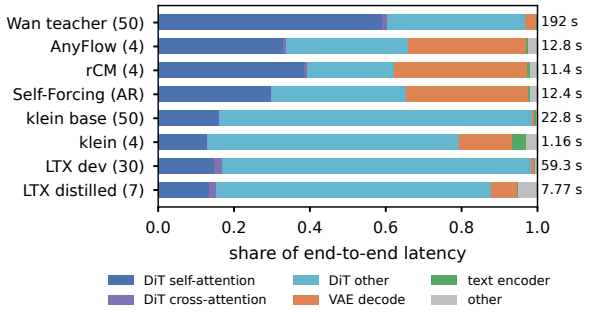


Figure 2. Where one generation spends its time, as released (latency probe; end-to-end times at right as in Tab. 4). After distillation, the VAE decoder and the non-attention DiT layers dominate.

4.3. The system caps the gain

A student makes far fewer DiT calls than its teacher, so components that were negligible for the teacher decide what an accelerator can deliver (Tab. 4 and Fig. 2). *The decoder* takes 31–35% of end-to-end time in the 1.3B Wan students, so even an infinitely fast DiT gives at most 2.6–3.4×; rCM at 14B spends 9.5% there and LTX, with its 32×-compressed latent, 6.8%. *The kernel's speed depends on shape*: block-sparse attention is 3.1× faster than dense for the Wan family, and no faster, or slower, for LTX, klein and Self-Forcing's per-chunk attention, which is short or already cheap. With a self-attention share of 13–16%, this caps sparse attention on LTX and klein at a few percent, whatever its quality. *INT8 follows model size, not distillation*: compiled W8A8 gains at most 1.04× at 1.3B, 1.13× at 14B and 1.23–1.24× on LTX, teacher and student alike, against our prediction ($< 1.15\times$). *The GPU decides even more*: on a B200, rCM's DiT runs 5.6× faster than on the A800 but its decoder only 3.1×, so the decoder takes 39–47% of a 1.3B student's latency, and the block-sparse kernel is no faster than dense at the Wan shapes.

		skipped call k						concordant pairs			
Student		1	2	3	4	5	6	S1	S2	S3	S4
AnyFlow	LPIPS	.210	.078	.074				3/3	2/3	3/3	1/3
	Δaes.	−.04	−.01	−.**10**							
rCM	LPIPS	.359	.283	.575				1/3	1/3	1/3	3/3
	Δaes.	+.01	+.**08**	−**2.00**							
klein	LPIPS	.225	.106	.092				3/3	3/3	3/3	1/3
	Δaes.	−.04	+.01	+.05							
LTX	LPIPS	.276	.143	.132	.079	.024	.009	8/15	6/15	14/15	13/15
	Δaes.	+.02	+.01	+.01	−.02	+.**09**	+.**08**				

Table 5. Fidelity cost (LPIPS to the unaccelerated output; 95% CIs are ±0.003–0.04) and quality effect (paired Δ aesthetic score; bold: 95% CI excludes 0) of skipping one call. Right: position pairs each cheap signal orders as the fidelity cost does.

5. Teacher Calibrations Mislead; a Short Measurement Does Not

5.1. Where a student can skip

We measure two costs of skipping call k, i.e. reusing the previous call's block-stack residual there. The *fidelity cost* is the LPIPS to the unaccelerated output of the same prompt and seed; caches are designed to minimise it. The *quality effect* is the paired change of reference-free scores, mainly aesthetics and DINO subject consistency. We relate the fidelity cost to four cheap signals (Tab. 5 and Fig. 3): S1, $1-$ the block-residual cosine; S2, the first block's relative residual change, i.e. FBCache's signal; S3, $1-$ the output's cosine to the previous call's; and S4, the output change caused by Gaussian noise of norm $0.1\|v_k\|$ added at call k. A signal *ranks* the positions if it orders at least 8 of the 9 position pairs of the three 4-step students and 13 of the 15 LTX pairs as the fidelity cost does (registered thresholds).

No cheap signal ranks skip positions across students. Each signal fails on at least one student. FBCache's signal is *largest* at LTX calls 4–5, among the positions with the lowest fidelity cost. In a registered comparison on held-out prompts, FBCache and TeaCache, tuned on the student to skip one call per video, never beat a measured skip. One skip is the natural unit with 4–7 calls, but we did not test more. FBCache skips LTX call 3 (LPIPS 0.117, where call 6 is below the floor) and rCM's last call (aesthetic -0.50, also at 14B). TeaCache's signal depends only on the timestep, so on a fixed sampler it is a fixed schedule; it skips AnyFlow call 1 (2.8× the fidelity cost).

Fidelity cost and quality effect are different quantities. They agree on rCM: its most expensive skip, the last call, also destroys the output (aesthetic -2.00, DINO -0.20; Fig. 4). On LTX, no skip lowers quality significantly, and early skips yield a different video of the same quality; this holds on held-out prompts and seeds. On AnyFlow they disagree: skipping the last call has the lowest fidelity cost (0.074) but is the only skip that lowers aesthetics (-0.10;

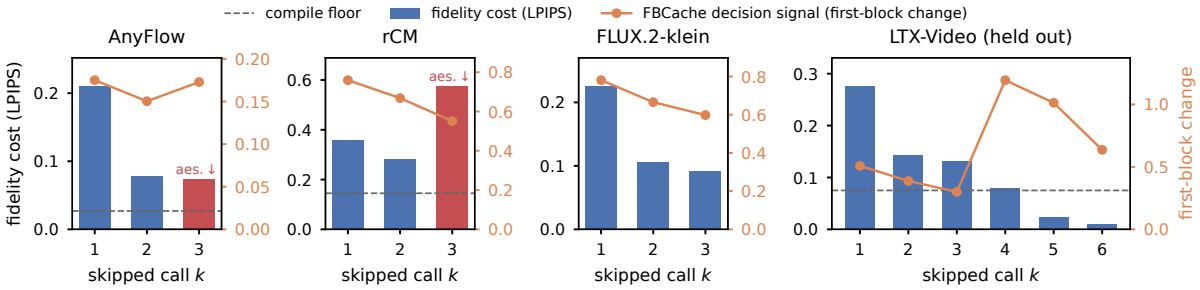


Figure 3. Fidelity cost of skipping one network call (bars; LPIPS to the student's unaccelerated output) and FBCache's decision signal at that call (line); both axes start at zero. Red bars mark the skips that significantly lower the aesthetic score (Tab. 5). The cache signal tracks the fidelity cost on klein but inverts it on the held-out LTX student and on rCM.

-0.15 [-0.20, -0.10] held out), while call 2 costs about the same (0.078) and leaves them unchanged. Late skips are thus nearly free in fidelity for the deterministic flow students, but not always in quality. These verdicts replicate on a B200 (7 of 7 effects, 13 of 14 nulls), and VBench agrees: rCM's last skip lowers imaging quality by 0.13, while AnyFlow's call 2 and every LTX skip lower no dimension.

5.2. Sparse attention

Making the attention sparse at exactly one call, for calls 0 to 3, costs LPIPS 0.430, 0.136, 0.047 and 0.023 on AnyFlow, so keeping the first call dense, as base-model sparse methods do, remains the right rule. With that rule, block-sparse attention (pooled-score top-k, density 0.2) and dropping one call give about the same measured speedup (1.18–1.22× vs 1.18×). Sparse attention drifts further: LPIPS 0.163 vs 0.081 on AnyFlow ($+0.082$ [0.074, 0.089]) and 0.352 vs 0.221 on rCM ($+0.131$ [0.119, 0.142]), replicated on held-out prompts ($+0.082$, $+0.118$) and larger at 14B ($+0.176$). The gap holds at every density we tried (0.1–0.5, one or two dense calls): settings as fast as one call fewer drift further, and those that drift no more are slower, since even an ideal kernel would save them at most 0.82 of a call. On LTX, sparse attention gives 1.05× at LPIPS 0.335, reusing the last call 1.16× at 0.009.

Mass kept does not decide the gap, and how keys are grouped does. At density 0.2 and $t \approx 940$, the attention mass kept is 0.93 for a per-token oracle, 0.85 for a per-block oracle and 0.78 for pooled scoring, for teacher and student alike. We generated held-out videos with each (registered; AnyFlow, rCM). An oracle choice of position blocks barely moves the drift (0.151 vs pooled 0.161 on AnyFlow, and worse on rCM, 0.353 vs 0.328), so it still loses to one call fewer (0.079, 0.218). An oracle choice of tokens keeps the most mass and drifts the most (0.30, 0.59 on 12 prompts), against our prediction; oracle coverage bounds mass, not drift. Our port of SVG2's semantic-aware permutation [33] (k-means on queries and keys, attention exact on the kept clusters) behaves differently: at density 0.23 it ties one call fewer on AnyFlow ($+0.002$ [-0.001, 0.004]) and beats it on rCM (-0.017 [-0.026, -0.009]), and at density 0.19 it loses ($+0.026$, $+0.021$). These speeds assume an ideal kernel and free clustering (1.19×, 1.23× vs 1.18×, 1.16×); its real speed on students is unmeasured, and it stays behind one call fewer on quality (DINO on both, aesthetics on AnyFlow). So the comparison above holds for selection over position blocks, not for sparse attention in general.

5.3. Autoregressive students

If context anchors later chunks, errors in the first chunk should propagate through the KV cache, and later chunks should tolerate fewer steps. On Self-Forcing (7 chunks, 4 calls per chunk), the drift from the 4-step output is 0.312 LPIPS for uniform 3 steps (21 calls), 0.131 for 4 steps in the first chunk and 3 in the rest (22 calls), and 0.155 for 4 and 2 (16 calls). Uniform 2 steps (14 calls) drifts by 0.450 and loses DINO consistency (-0.013; held out, -0.008); the anchored 16-call schedule does not. The gaps replicate on held-out prompts (-0.178, -0.151 against uniform 3). Part of the gap is the anchoring itself: the first chunk is computed as in the reference, so later chunks attend to the same context. This gives a mechanism and a size to the "first chunk full, rest reduced" schedules [20].

5.4. The protocol

An 8-prompt sweep. Students have only 4–7 calls, so both costs of each skip position can simply be measured. On 8 tuning prompts, disjoint from the evaluation prompts, a registered check orders 22 of 23 position pairs as the 33-prompt development set does and finds the same cheapest position on all four students, for 0.6–6.9 GPU-minutes per student. Quality needs more care: the sweep flags rCM's last skip, but aesthetic changes of 0.1 are within its noise. We therefore recommend choosing positions by fidelity on a few prompts, confirming quality on 30 or more, and moving to the next-cheapest position if quality drops. On AnyFlow this replaces call 3 by call 2. At the held-out 14B scale the sweep again picks call 2, and the last call again breaks the



Figure 4. Frames at 25% and 75% of the clip, for the prompt whose LTX skip-1 fidelity cost is closest to the median (not selected by appearance). LTX skip 1 has a high fidelity cost (a slightly different video) but no quality loss. rCM skip 3 breaks the output.

Student	Speedup	Pred.	LPIPS	Δaes.	ΔDINO	Δmotion	GB
Wan2.2 A14B	2.69×	2.67×	.414	−.04	−.002	**−16%**	+9
AnyFlow	3.45×	3.46×	.256	**−.10**	**−.007**	**−22%**	+13
rCM	3.36×	3.33×	.288	+.02	.000	**−9%**	+13
rCM-14B (held out)	2.53×	2.50×	.260	−.06	**−.004**	**−8%**	+13

Table 6. Two calls + tiny decoder + compile on a PRO6000 BW, 66 held-out videos, against each student's base. Pred.: from separately timed parts. Bold: 95% CI excludes 0. Motion relative to the base's level. VBench's three scores do not fall on any.

output (aesthetic -1.42). On two third-party LightX2V students, the pick needs the quality check on Wan2.2 A14B (last call: aesthetic -0.08) and ties on HunyuanVideo-1.5. Run-time adaptivity has little room: a per-video oracle gains 2.6% over the best fixed two-call grid on Wan2.2 (registered) and, in an exploratory count, 0–8% over the best fixed skip on six students.

A latency-split probe. Amdahl's law, $1/((1-f)+f/s)$ for latency share f and speedup s, is not new; we test whether the probe measures f and s well enough to apply it before running a method. On the held-out LTX pair all predictions fell within 1.3–4.9% of the measurements (registered bound: 15%), at the held-out 14B scale within 1.4%, and for 16 sparse runs on an H100 within 0.8%. On a B200 it predicted step runs within 0.4% and Wan sparse runs within 3%, but overstated the sparse slowdown at LTX's short attention (16.5%, 25.0%).

6. A Baseline to Beat

Removing costs outside the DiT. A tiny trained replacement decoder (`lightx2v/Autoencoders`, as released) lifts the decoder's cap: on an H100, 1.50× and 1.63× on AnyFlow and rCM at LPIPS 0.028 and 0.037, and 1.18× on Wan2.2 A14B. Against our prediction imaging quality does not fall; DINO falls by at most 0.007, rCM's aesthetics by 0.047, and peak memory rises from 18 to 31 GB. With one skip and compile it reaches 2.47× at LPIPS 0.097 on AnyFlow, within 4.8% of the prediction.

Two calls, a tiny decoder, compile. Spending the calls where the grid says and removing the costs outside the DiT reaches 2.5–3.5× end to end, each within 1.0% of a prediction from separately timed parts (Tab. 6). Only rCM-14B is a held-out test; on the others the parts' costs were already known. Drift stays within 14% of the two-call grid's. It is not free: motion falls on all four, DINO on two and aesthetics on AnyFlow, about as for the grid alone, and on top of the grid decoder and compile lower aesthetics on three of the four (by 0.05–0.14), against our prediction. Whether a lower motion score is a visible loss we have not tested. This combines released components and is not a method; we offer it as what a training-free student accelerator should beat.

Frame sparsity and allocation do not beat the grid. Our reimplementation of FIS-DiT [24] (H100, Wan2.2 A14B; residual reading, as the literal one gives speckled frames) replicates its quality claim: aesthetics, DINO and VBench's three scores do not fall, though motion does (-1.88 on 4.23). At about equal speed the two-call grid drifts less (-0.087 [-0.103, -0.070]) and, by aesthetics and DINO, is quality-neutral on this student (motion -0.58). With decoder and compile on both, at 2.63× vs 2.69×, FIS still drifts further ($+0.081$ [0.063, 0.098]) and loses more motion (-1.90). Among frame-sparse plans, a dense first call drifts less than a dense last one on all three students tested (-0.062, -0.240, -0.208), as amplification predicts, but lowers VBench's scores (7 of 9 CIs below 0): drift and quality ask for different calls. A planner allocating frame sparsity and reuse from single-call costs predicts speed within 2.1% but beats the grid on no student, so by the registered rule we make no method claim (supplement).

7. Pre-registered Predictions and Outcomes

After studying Wan and klein, we committed predictions, thresholds and decision rules before any held-out run and later diagnostic; Tab. 7 lists every outcome. All later diagnostics came after earlier results, each registered before its own runs, so the table marks which predictions met new data.

	Prediction (LTX unless noted)	Outcome
H	P1 Student redundancy equals the teacher's over the same interval ($\lvert\Delta\rvert \leq 0.20$); tiny early steps redundant ($\cos \geq 0.90$)	**failed**: first part held (≤ 0.073), second not (0.69–0.86)
H	P3 Early skips cost $\leq 2\times$ the compile floor, late ones $\geq 3\times$ the early median	not tested as P1 failed; measured: the *opposite* order
H	P4 Attention coverage, student vs teacher at matched t: $\lvert\Delta\rvert \leq 0.03$	**held** ($\lvert\Delta\rvert \leq 0.018$)
H	P5 Amdahl within 15%; sparse attention $< 1.10\times$ if self-attention $< 20\%$; INT8 DiT $< 1.15\times$	**held** (1.3–4.9%); **held**; **failed** ($1.23\times$, $1.24\times$)
H	P6 Headline differences replicate held out (Wan)	**held** (4/4)
E	D2 Late skips are cheap due to the decoder / local error / amplification	rejected / rejected / **supported**
S	D3 Some cheap signal ranks skip positions on every student	**failed**: none does
S	D4 8 prompts reproduce the ranking ($\geq 21/23$ pairs, same cheapest)	**held** (22/23, 4/4)
H	D5 Held-out: AnyFlow's call-3 skip lowers aesthetics, call 2's does not	**held** (-0.15; -0.02 n.s.)
H	D6 Held-out: LTX's ranking reproduces ($\geq 13/15$), no skip lowers quality, calls 5–6 raise aesthetics	**held** (15/15; $+0.08$, $+0.06$)
H	D7 Student-tuned FBCache vs protocol (LTX / rCM / AnyFlow); D8 rCM: reuse drops more	**held / failed / failed**; **held**
H	D9 rCM-14B: VAE $\leq 15\%$, INT8 $\geq 1.10\times$, kernel $\geq 2.5\times$, probe, skips, FBCache, sparse vs drop	**held** (9.5%, $1.13\times$, $3.15\times$, $\leq 1.4\%$, 2/2, -1.42, 8/8, $+0.18$)
H	D10 14B pair: coverage $\lvert\Delta\rvert \leq 0.03$; interval explains most lost redundancy	**held** (≤ 0.014; 55%)
H	D11 B200: probe within 15%; VAE $\geq 20\%$; kernel $< 2\times$; INT8 follows size	**failed** (LTX sparse); **held** (3/3)
S	D13 Equal key count: coverage rises with AR context; Wan above LTX	**failed**; **failed** (key count)
S	D14 Densities 0.1–0.5: equal-speed sparse drifts more; a slower one less; probe; equal drift saves < 1 call	**held** (4/4)
H	D15 B200 replicates quality verdicts; VBench agrees	**held** (4/4; 1 flip allowed)
H	D16 Third-party Wan2.2, HunyuanVideo-1.5: VAE, probe, order, pick, quality, FBCache, sparse	8/12 **held**, 4 **failed** (supplement)
S	D17 FIS (ours) vs two calls, its quality, probe, decoder, stacks; two calls, decoder cost quality	6/8 **held**; 2 **failed** (both cheaper)
S	D18 Allocating calls: adaptivity, dense first, predictions, stack; quadrature; planner beats grid	4/6 **held**; 2 **failed** (2/3, 3/3 models)
S	D19 Two calls + decoder + compile: probe, $\geq 2.5\times$, drift, vs FIS; no quality cost beyond the grid (rCM-14B: H)	4/5 **held**; **failed** (2/3 models)
S	D20 Density 0.2: oracle blocks lose to one call fewer; oracle tokens beat it; oracle blocks beat pooled; SVG2-style port (ours) loses	**held**; **failed**; **failed** (rCM); **failed** (at $p = 0.9$)
S	D21 Same states: interval share $\geq 50\%$, teacher closer than one step; early errors amplify on five students; coverage stable	**held** (5/5)

Table 7. Pre-registered predictions and outcomes. H: new data (family, scale, prompts, GPU or third-party students). S: registered on models already studied. E: registered diagnostic after a surprise. A further, out-of-scope prediction and the record are in the supplement.

What the failures changed. Twenty predictions failed; each narrowed a claim, and none tested attention concentration or the interval's share, which hold on new data and on the same states. The step axis is not about interval width (P1); no cheap signal ranks skip positions, so we measure skip costs (D3). INT8 was extrapolated from 1.3B and was wrong at 13B (P5); FBCache's skip does not act like plain reuse (D7, D8); the kernel benchmark misjudges very short attention on a B200 (D11); coverage differences come from the key count (D13). Two calls and the tiny decoder cost less quality than we assumed (D17), allocation took away a method claim we had set out to make (D18), the baseline is not free (D19), and the mass an attention selection keeps does not decide its drift, which removed our account of why sparse attention loses (D20). Six P6 comparisons, on a question that left the paper's scope, were withdrawn before any held-out comparison was computed (supplement); the four kept replicate.

8. Limitations and Conclusion

Limitations. Timings come from four GPU types (supplement); kernel and INT8 numbers depend on the GPU, sparse attention is timed with one kernel, and FIS and the SVG2-style selector are our own ports, the latter without its kernel. rCM's checkpoints are an unofficial reproduction; its last-skip failure rests on them. AR students are tested on 5-s videos. Quality rests on 33–66 prompts per model, reference-free scores and frames, not on a user study; fidelity is read only above the compile floor; we cover three families (and two more in part).

Conclusion. Training-free accelerators carry their teacher's calibration to students whose features have changed less than one of their own steps, but whose grid (which amplifies early errors) and system (which caps the gain) have changed. Both can be measured on the student in minutes, and pre-registered held-out tests largely confirm them. Acting on both gives a 2.5–3.5× baseline at stated quality costs.

LLM Usage Statement

This work was carried out with extensive use of Anthropic's Claude (Opus models), run as an autonomous coding agent through Claude Code. Under the authors' direction, the agent searched and summarised the literature, proposed candidate research questions, wrote the code of the measurement harness, probes, accelerators and analysis scripts, launched and monitored the experiments, drafted the pre-registrations and their scoring scripts, analysed the results, produced the figures and tables, and drafted and edited the text of this paper and its supplement. The authors chose the research question, the venue and the scope, decided which experiments to run and on which compute, and take full responsibility for the content. Every reported number comes from the project's scripts and logged runs, and the registration record (supplement) lists the commits that fixed each prediction before its runs.

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